

How high school students could find new theorems using computer algebra systems.

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I and high school students have been doing a research of mathematics and have found a dozen new theorems of mathematics, and some of them have been already published as papers or been introduced at international conferences. Even high school students can understand these theorems, but even professional mathematicians find them new and interesting.

At ACA 2005 we are going to introduce "General theory of Russian roulette" and other researches of us.

Let us introduce a part of our research.

In this generalized Russian roulette game players take turns and pull a trigger to themselves. We denote by $F[p,n,m,v]$ the probability of death of the v -th player in game with p players, n cylinders and m bullets. If we make a pyramid shape of $F[p,n,m,v]$ when $p = 3$ and $v = 2$, then we get a very interesting pyramid of fractions.

$$\begin{array}{c}
 \{F[3, 1, 1, 2]\} \\
 \{F[3, 2, 1, 2], F[3, 2, 2, 2]\} \\
 \{F[3, 3, 1, 2], F[3, 3, 2, 2], F[3, 3, 3, 2]\} \\
 \{F[3, 4, 1, 2], F[3, 4, 2, 2], F[3, 4, 3, 2], F[3, 4, 4, 2]\} \\
 \{F[3, 5, 1, 2], F[3, 5, 2, 2], F[3, 5, 3, 2], F[3, 5, 4, 2], F[3, 5, 5, 2]\}
 \end{array}$$

Note that the following pyramid-like shape has an interesting structure. It is very much like a Pascal's triangle, but it has a double structure.

$$\begin{array}{c}
 \{\frac{0}{1}\} \\
 \{\frac{1}{2}, \frac{0}{1}\} \\
 \{\frac{1}{3}, \frac{1}{3}, \frac{0}{1}\} \\
 \{\frac{1}{4}, \frac{2}{6}, \frac{1}{4}, \frac{0}{1}\} \\
 \{\frac{2}{5}, \frac{3}{10}, \frac{3}{10}, \frac{1}{5}, \frac{0}{1}\}
 \end{array}$$

Now I am studying the method of research with high school students under the guidance of professor Tadashi Takahashi who is a committee member of ACA 2005. More than 30 students contributed to our research, and I am sure that the research is within the range of high school students who love mathematics if they can use a good computer algebra systems.